

Industrial Internship Report on “Smart Agriculture: Crop & Weed Detection using Deep Learning (YOLOv8)”

**Prepared by
Meghram Bairwa**

Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT, and we had to finish the project including the report in 6 weeks' time.

My project was focused on developing a Computer Vision-based AI model that can automatically detect and differentiate between crops and weeds in agricultural fields. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement a solution for that. It was an overall great experience to have this internship.

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1 Preface

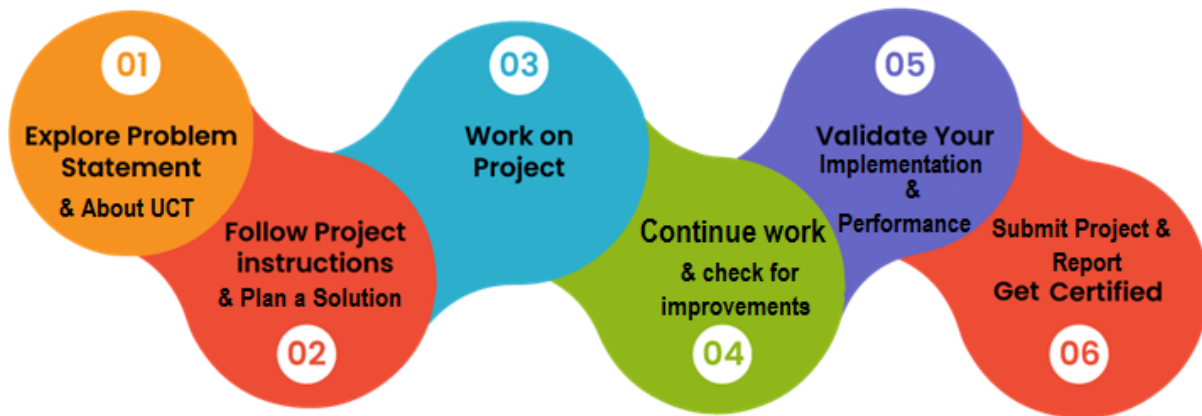
During this 6-week internship, I successfully developed an AI-powered Crop & Weed detection system. In today's tech-driven world, a relevant internship is crucial for career development as it bridges the gap between theoretical academic knowledge and practical industry applications.

The opportunity given by USC/UCT allowed me to dive deep into Deep Learning, specifically working with the YOLOv8 algorithm and Streamlit for web deployment. The program was well-planned, starting from exploring the problem statement, planning a solution, working on the model, validating its performance, and finally submitting the complete project.

I would like to thank my mentors at UCT and my peers (Ankur, Neeshoo, Sudhanshu, and Mohit) who directly or indirectly helped me during technical brainstorming sessions.

Message to juniors: Always focus on the constraints of a real-world dataset. A model is only as smart as the data you feed it!

How Program was planned



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



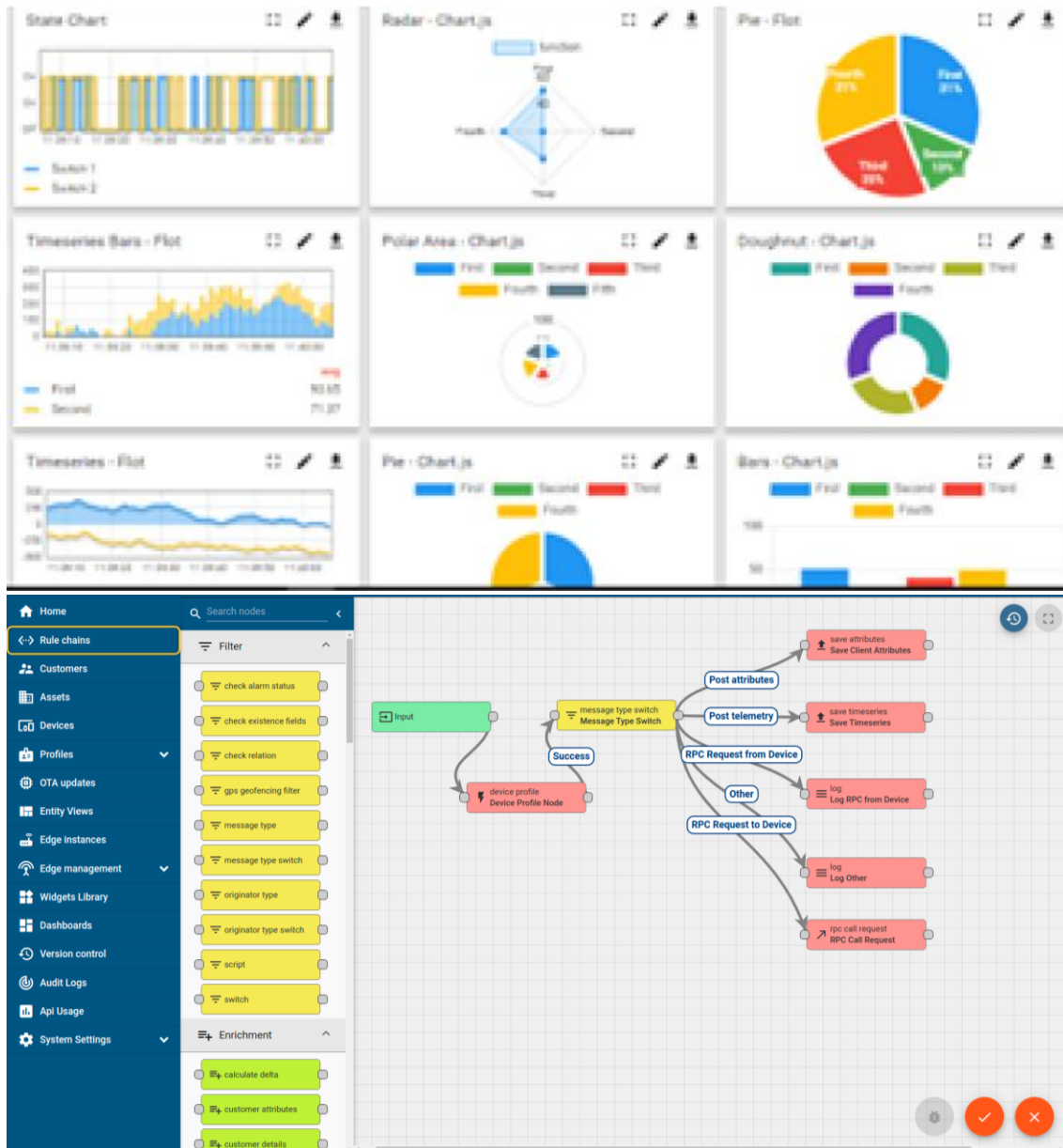
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



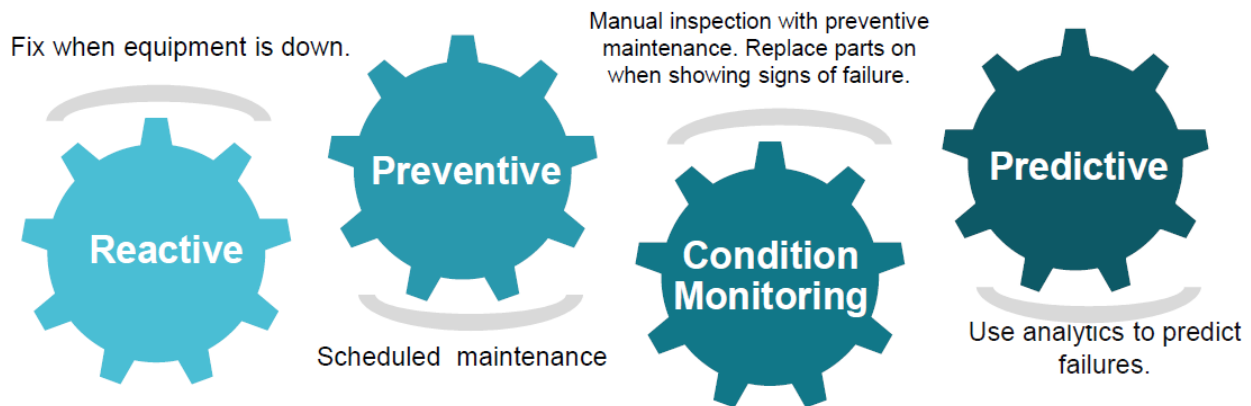


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

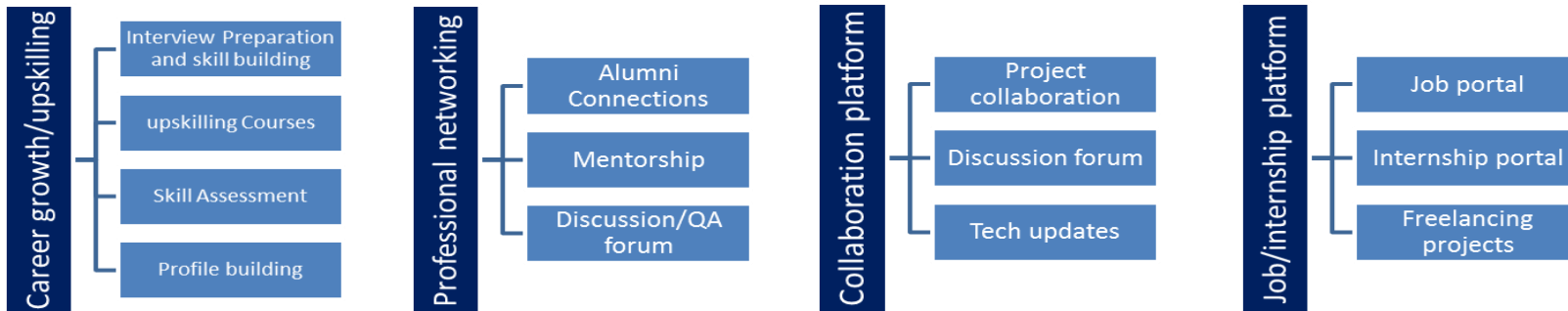
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Ultralytics YOLOv8 Official Documentation
- [2] Streamlit Python Library Documentation
- [3] OpenCV for Computer Vision

2.6 Glossary

Terms	Acronym
AI	Artificial Intelligence
YOLO	You Only Look Once (Object Detection Model)
IoU	Intersection over Union
SaaS	Software as a Service

3 Problem Statement

In the assigned problem statement

Weeds consume essential nutrients, water, and sunlight that are primarily meant for agricultural crops, leading to a significant reduction in overall farm production. In modern agriculture, identifying weeds rapidly and accurately is crucial for automated pesticide spraying and smart farming, saving crops from being damaged by toxic chemical over-usage.

4 Existing and Proposed solution

- **Existing Solution:** Traditional farming relies on broad-spraying of pesticides.
- **Limitations:** This method wastes highly expensive chemicals, damages the environment, and harms the main crop's health.
- **Proposed Solution:** I propose an AI-based Computer Vision system utilizing the YOLOv8 object detection model. This system scans field images and draws bounding boxes to precisely classify "Crop" and "Weed".
- **Value Addition:** I developed a user-friendly Web Interface using Streamlit so that any user can upload multiple farm images and get real-time AI predictions without writing a single line of code.

4.1 Code submission (Github link): [Click Here For Get Code File](#)

4.2 Report submission (Github link): [Click Here For Get Report File](#)

5 Proposed Design/ Model

- **Foundation & Start:** Objectives defined: Precision detection (<50ms MIPS constraint) to reduce chemical waste. Constraints: Domain robustness (Textures: Spiky vs. Smooth Leaves).
- **Initial Build:** Trained the YOLOv8 model (best.pt) on Dataset 1 (Binary data: smooth broad-leaf species).
- **Iterative Testing:** Subjected the initial PoC (best.pt) to stress testing using wide-angle and novel unseen species (Cotton, Wheat, Sugarcane).
- **Failure Analysis Node:** This crucial intermediate stage (as analyzed in images 9-12) detected severe misclassifications: Domain Shift/Texture Bias. The model interpreted spiky textures (cereal crops like Wheat) as "Weeds," proving the data was "Out of Distribution" (OOD).
- **Optimization & Validation:** Constraints validated: Speed constraint (avg 3.5ms) was met. Thresholding was re-adjusted (0.20 -> 0.50). Future scope defined: Dataset 2 update with spiky cereal crop data for diverse multi-class robustness.
- **Final Outcome:** Successful delivery of a Streamlit Web Interface (App) running the optimized, low-latency PoC for real-time field analysis, with a defined path for industrial scaling (integrating into UCT's Factory Watch ecosystem).
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5.1 High Level Diagram (if applicable)

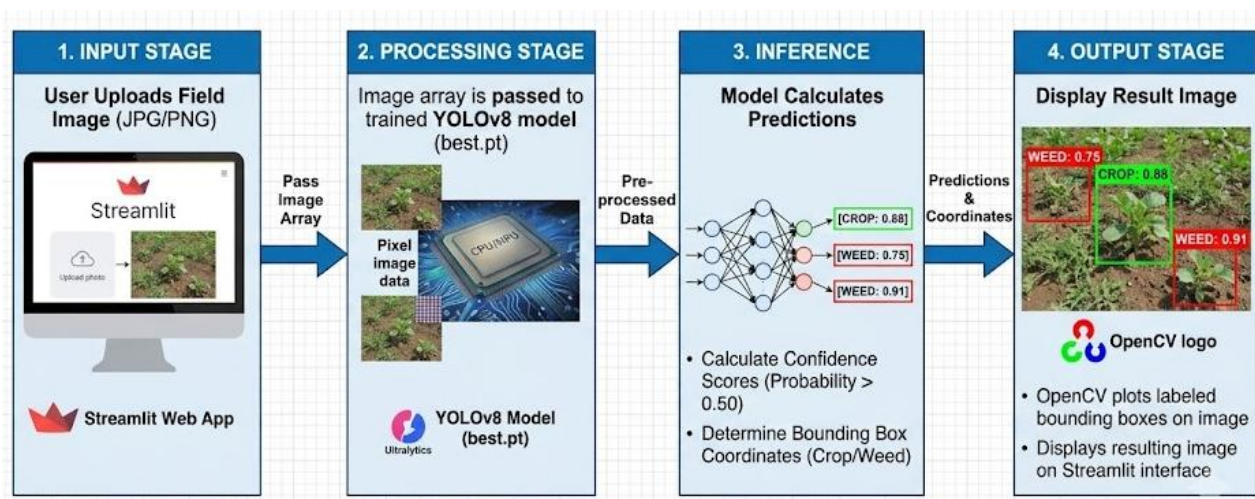


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

5.3 Interfaces (if applicable)

Smart Agriculture: Crop & Weed Detection

Upload images of a field to automatically detect crops and weeds using AI.

Choose images (JPG, JPEG, PNG)...



Results for: img2.jpg



Results for: img1.jpg



Detection Result: img1.jpg

 All Detections Complete!

6 Performance Test

For this project, testing was done keeping industrial constraints in mind: high processing speed (MIPS) and varying environmental backgrounds.

6.1 Test Plan/ Test Cases:

- Tested the AI against wide-angle shots, close-up shots, and entirely unseen crop species (like Wheat and Cotton) to evaluate Domain Shift and model robustness.

6.2 Test Procedure:

- Images were passed through the predict function with a strict confidence threshold set to 50% (conf=0.5).

6.3 Performance Outcome:

- Speed Constraint: The model performed exceptionally well, achieving inference times of just 3.5ms to 45ms per image.
- Accuracy Constraint: Successfully identified broadleaf weeds and crops with high confidence (60%-80%).
- Limitations Identified: The model showed False Positives for previously unseen spiky crops (like Wheat), proving that the model is actively analyzing textures but requires diverse multi-class data to avoid "Out of Distribution" failures.

7 My learnings

This project provided immense technical and logical growth. Key learnings include:

1. **Deep Learning Pipeline:** Mastered data augmentation, annotation (YOLO format), and training neural networks for custom datasets.
2. **Real-world AI Constraints:** Understood practical challenges like *Data Bias* and *Domain Shift* (e.g., the model confusing a flower or cotton for a weed because it wasn't in the training vocabulary).
3. **Deployment:** Learned to bridge the gap between backend AI logic and front-end user experience using Streamlit and OpenCV.

8 Future work scope

Due to time limitations, the model was trained on a binary class system (Crop vs. Weed) using primarily broadleaf datasets. In the future, the scope can be expanded by:

- Adding more specific crop classes (e.g., Wheat, Cotton, Sugarcane) to eliminate false positives.
- Introducing a third class for "Ignored Objects" like common field flowers.
- Integrating the AI model with UCT Insight platform or drone cameras for real-time video stream detection in actual agricultural fields.